



Part 10

The Art of Guard Macros

It feels good to return to the SELinux series after a break of about two months. What started initially as a small article on SELinux has gradually developed into a series running over 10 articles, and I feel a few more will be required for a proper introduction to the subject.

This month, let's talk about the powerful macros SELinux has, which can help you automate most of your rules.

We have covered security contexts, allow rules, type transitions for objects and subjects, and more importantly, creating our own SELinux modules as per our system requirements. The more policies you write, the more you realise that a lot of commands are repetitive. SELinux has powerful macros to automate most of your rules. You just need to look up the relevant macros and use them in your type enforcement files.

Even the example files provided along with the `selinux-policy` rpm (on my FC12 box) make extensive use of macros. Let us examine the `example.te` file provided as a template to create modules for an application (myapp):

```
[vbg@vbg-work devel]$ cat /usr/share/selinux/devel/
example.te

policy_module(myapp,1.0.0)
```

```
#####
##
## Declarations
##
#
type myapp_t;
type myapp_exec_t;
domain_type(myapp_t)
domain_entry_file(myapp_t, myapp_exec_t)
#
type myapp_log_t;
logging_log_file(myapp_log_t)
#
type myapp_tap_t;
files_tap_file(myapp_tap_t)
#
#####
##
## Myapp local policy
##
```